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GeoFGA: Fine-Grained Image–Text Alignment for Open-Vocabulary Remote Sensing Image Semantic Segmentation

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***Abstract*—Open-vocabulary remote sensing image semantic segmentation (OVRSIS) leverages pretrained vision–language models to align visual features with textual semantics in a shared embedding space, enabling pixel-level recognition of both seen and unseen classes. Despite this capability, existing methods often struggle to preserve reliable image–text alignment for dense prediction. Ambiguous class semantics and interference from adjacent land-cover regions further distort dense image–text correlations, leading to erroneous activations and class confusion. To address these issues, we propose GeoFGA, a fine-grained image–text alignment framework for OVRSIS. Specifically, we introduce a Geospatial Attribute-Guided Text Enhancement (GATE) module, which organizes LLM-generated category descriptions into a shared attribute prototype space and utilizes image-conditioned attribute composition to capture context-aware and discriminative geospatial semantics. Furthermore, a Hierarchical Correlation Recalibration (HCR) strategy is proposed to refine dense image–text correlations through spatial, feature, and category reliability recalibration with GATE-enhanced textual features, thereby enhancing adaptability in remote sensing scenarios. Extensive experiments on seven remote sensing datasets demonstrate that GeoFGA achieves the best average performance among the compared methods and improves the discrimination of unseen classes.**

***Index Terms*—Open-vocabulary remote sensing image semantic segmentation, vision-language models, CLIP, attribute prototypes, correlation recalibration.**

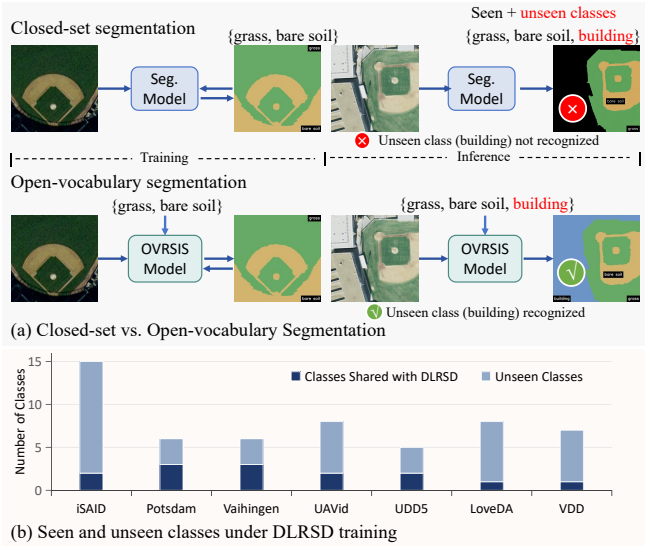


Fig. 1. An illustration of open-vocabulary remote sensing image semantic segmentation. Conventional segmentation models trained on closed-set classes (grass and bare soil) fail to recognize unseen classes (e.g., building), while OVRSIS models enable recognition of unseen classes beyond the training categories. (a) Comparison between closed-set and open-vocabulary segmentation. (b) Numbers of seen and unseen classes in the evaluation datasets, with DLRSD used for training.

I. Introduction

INSPIRED by recent advances in open-vocabulary learning [1], [2], open-vocabulary remote sensing image semantic segmentation (OVRSIS) aims to assign each pixel in a remote sensing image a semantic label from an open vocabulary, thereby reducing the reliance on predefined class sets. Unlike conventional closed-set semantic segmentation, OVRSIS leverages pretrained vision–language models [3] and textual class descriptions to recognize and segment both seen and unseen classes [4]. As illustrated in Fig. 1, this capability allows the class vocabulary to be flexibly defined at inference time while reducing the need for pixel-level annotations when introducing new classes.

Prevalent methods typically follow a correlation-based pipeline, including correlation map generation, refinement, and segmentation decoding [5]. First, correlation maps are computed by measuring similarities between dense image features and text features extracted by CLIP [3]. Since CLIP is pretrained for image-level vision-language alignment rather than dense prediction, the generated correlation maps are often coarse [6]. These maps are further refined through spatial or class aggregation [7]. Finally, the refined correlation maps are decoded into segmentation predictions. Recently, many studies have improved open-vocabulary dense prediction by enhancing visual or textual representations [8], [9]. Despite improved semantic matching between the two modalities, the resulting dense image–text correlations may still produce unreliable class responses, limiting accurate pixel-level discrimination.

A key challenge in OVRSIS lies in establishing reliable image–text correspondences in remote sensing scenes. In remote sensing images, adjacent regions belonging to different classes may exhibit similar visual appearances [10], while indistinct boundaries further introduce semantic ambiguity [11], making accurate pixel-level alignment difficult. As illustrated in Fig. 2, interference from adjacent land-cover objects and inter-class visual similarities induce image–text matching deviations, resulting in unreliable correspondence and consequently causing class confusion and localization errors in pixel-level predictions. Semantic enhancement methods im-

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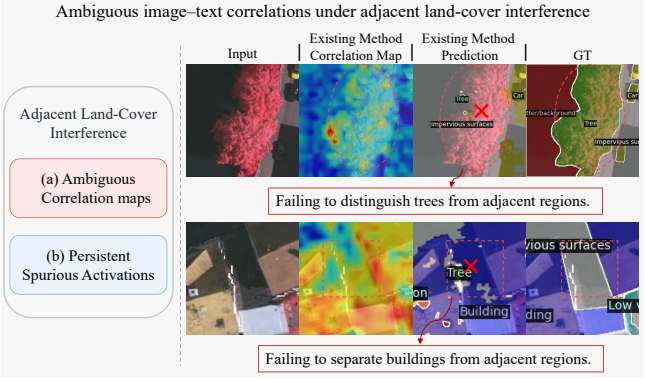


Fig. 2. Adjacent land-cover interference in OVRSIS. (a) Ambiguous initial image–text correlation maps. (b) Prediction errors caused by persistent spurious activations. The examples show that pixels belonging to trees and buildings are confused with those in neighboring non-target regions. GT denotes the ground truth.

prove the discriminability of category representations by introducing additional textual knowledge [8], [12]. However, these methods typically rely on class-level semantic descriptions to construct fixed textual representations for each category, making it difficult to characterize fine-grained semantic differences among visually similar categories and consequently limiting cross-modal alignment accuracy in complex scenarios. Correlation refinement methods improve dense image–text matching by aggregating contextual information within correlation maps [5], [13]. Nevertheless, improved contextual consistency does not necessarily ensure accurate image–text correspondence. In remote sensing scenes with strong inter-class visual similarities and neighborhood interference, ambiguous or spurious class responses may persist after refinement, leading to class confusion and inaccurate localization. Therefore, improving OVRSIS accuracy in remote sensing scenarios requires further enhancing the fine-grained semantic discriminability of textual representations and the reliability of dense image–text correspondences.

To address these limitations, we propose GeoFGA, a geospatial fine-grained alignment framework for OVRSIS. GeoFGA is built upon two complementary design principles: attribute-guided semantic representation and reliability-oriented correlation recalibration, aiming to improve semantic discrimination and image–text correspondence reliability in remote sensing scenes. The Geospatial Attribute-Guided Text Enhancement (GATE) module improves semantic discrimination among visually similar categories by composing geospatial attribute prototypes into image-adaptive textual representations. The Hierarchical Correlation Recalibration (HCR) module further enhances dense image–text correspondence by hierarchically recalibrating unreliable correlation responses. The main contributions are summarized as follows:

1) We propose a GATE module, which constructs a shared attribute prototype space and performs image-conditioned attribute composition to generate adaptive and discriminative semantic representations for improved image–text alignment.
2) We develop the HCR module to improve the reliability of dense image–text correspondences. HCR integrates spatial, feature, and category reliability recalibration guided by image context and attribute-enhanced textual representations. These complementary stages are designed to address spatial inconsistency, feature redundancy, and inter-class ambiguity, thereby suppressing unreliable matching responses.
3) Extensive experiments on seven remote sensing benchmark datasets demonstrate the effectiveness and generalization capability of GeoFGA. Visualization results further verify its ability to generate discriminative semantic responses and suppress erroneous activations in challenging scenarios.

II. RELATED WORK

A. Remote Sensing Image Segmentation

Remote sensing image segmentation has been extensively studied for dense land-cover mapping. Early approaches mainly rely on convolutional encoder–decoder architectures (e.g., FCN [14] and U-Net [15]) to learn hierarchical visual representations for pixel-level prediction. Recent studies further introduce Transformer-based architectures (e.g., AerialFormer [16]) to capture long-range contextual dependencies. Meanwhile, task-specific segmentation methods have been extensively developed to address particular application requirements. For instance, language-guided tasks, such as referring remote sensing image segmentation [17], incorporate textual modalities through vision-language interaction to facilitate target localization. Similarly, specialized network designs have been proposed for fine-grained target extraction, such as road extraction [18], [19]. Despite these advances, most existing methods still follow a closed-set paradigm where semantic categories or target objectives are fixed during training. This limitation restricts their ability to recognize novel categories beyond the predefined task scope.

B. Open-Vocabulary Semantic Segmentation

Open-vocabulary semantic segmentation (OVSS) extends semantic segmentation beyond a fixed label space by defining the class vocabulary through text. With the strong image–text alignment capability of vision–language models, particularly CLIP [3], many studies have explored extending open-vocabulary recognition to dense prediction [20], [21], [22]. Existing OVSS approaches generally follow a text–visual alignment paradigm, in which textual embeddings are matched with dense visual features to produce pixel-level predictions [23], [24], [25].

Subsequent studies have improved this paradigm through stronger textual guidance and more fine-grained dense matching. For example, CLIPSeg [26] enables prompt-driven segmentation, FreeSeg [27] introduces adaptive prompt learning for unified open-vocabulary segmentation, SCAN [28] employs semantic-assisted calibration to improve image–text alignment, and SED [29] constructs hierarchical cost maps for efficient dense prediction. More directly, CAT-Seg [7] formulates dense image–text similarities as cost volumes and

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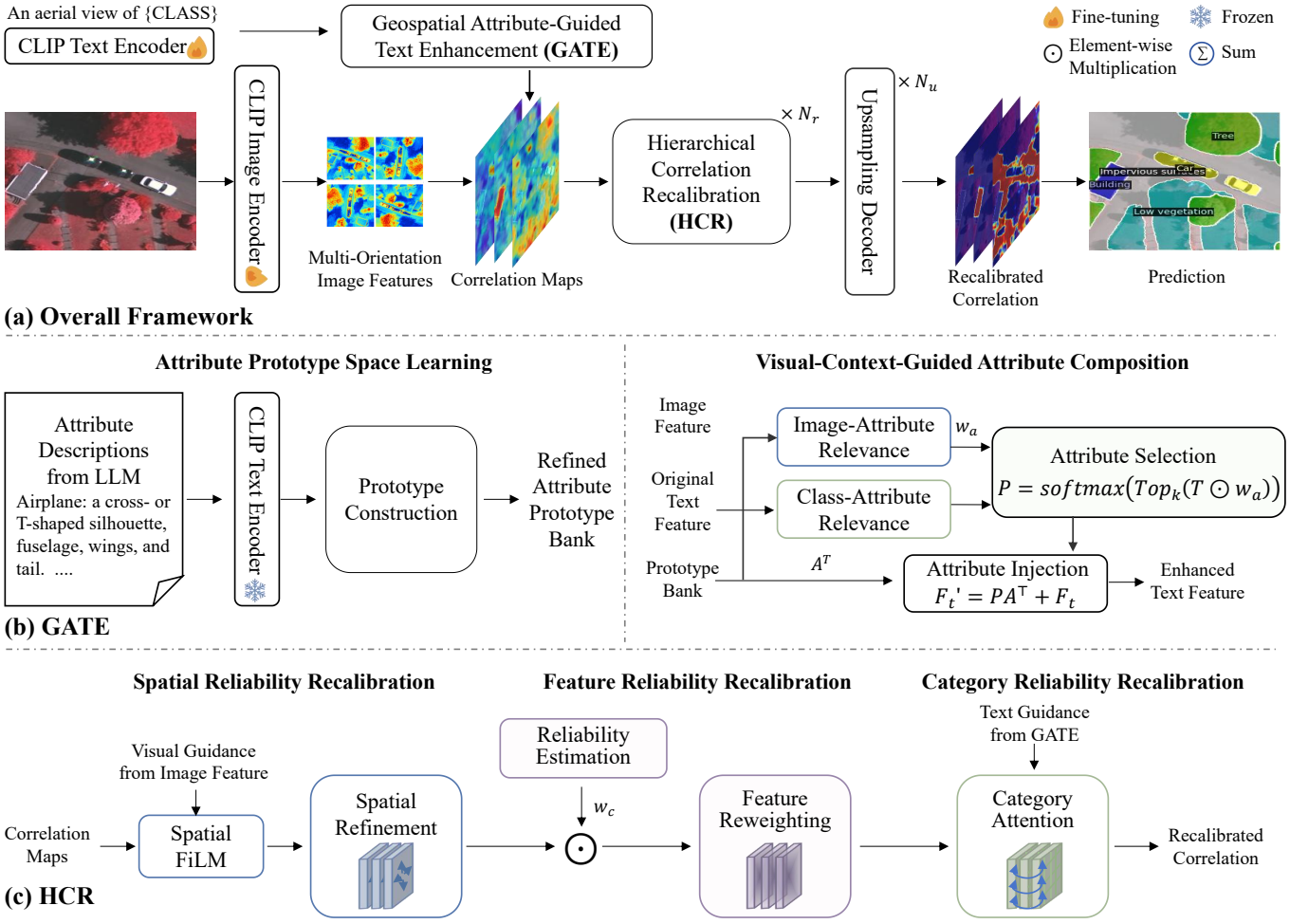


Fig. 3. Overview of the proposed GeoFGA. (a) Overall framework. GATE enhances text features by composing image-adaptive geospatial attributes. The enhanced text features are then matched with multi-orientation image features to construct image–text correlation maps. HCR hierarchically recalibrates these correlations before segmentation decoding. (b) Illustration of GATE. LLM-generated attribute descriptions are encoded and organized into a shared attribute prototype space. Image-conditioned attribute composition selects relevant attribute prototypes and integrates them into the text features. (c) Illustration of HCR. The image–text correlations are hierarchically recalibrated along the spatial, feature, and category dimensions to produce refined correlations for segmentation prediction.

refines them through spatial and class aggregation, while PCA-Seg [30] further explores parallel cost aggregation to better preserve complementary spatial and semantic information. Although these methods have substantially advanced open-vocabulary dense prediction, most are developed for natural-image scenarios and do not explicitly account for the distinctive visual and spatial characteristics of remote sensing images.

C. Open-Vocabulary Remote Sensing Image Semantic Segmentation

Building on advances in OVSS, recent studies have extended open-vocabulary dense prediction to remote sensing images, enabling the recognition and segmentation of land-cover classes beyond fixed label sets [31], [32], [33]. However, remote sensing images exhibit distinctive characteristics, including arbitrary object orientations, large scale variations, repetitive textures, and complex spatial layouts, which make the direct transfer of pretrained vision–language models challenging.

To address this challenge, existing OVRSIS methods have explored various strategies to adapt these pretrained models to complex aerial scenarios. On the visual side, while several studies focus on enhancing visual representations or incorporating domain-specific priors [5], [13], [34], they primarily concentrate on visual feature alignment. Consequently, the semantic ambiguity of category representations and unreliable image–text correspondences remain insufficiently addressed. On the textual side, recent methods enrich category representations by incorporating additional semantic knowledge, such as remote-sensing-specific descriptions or structured semantic relations [8], [12]. Nevertheless, these techniques typically rely on predefined semantic descriptions. Without explicitly modeling fine-grained attribute–category relationships, they struggle to capture context-dependent geospatial semantics. Beyond semantic enhancement, several methods refine dense image–text correlation maps by modeling contextual dependencies across spatial locations and categories [5], [8], [13]. Nevertheless, neighborhood interference, feature redundancy, and inter-class semantic ambiguity can still lead to unreliable

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matching responses in complex remote sensing scenes, posing challenges to accurate pixel-level image–text alignment.

III. Methodology

The overall framework of GeoFGA is illustrated in Fig. 3(a). Given a remote sensing image and textual class descriptions, GeoFGA extracts image and text features with CLIP and improves image–text alignment through geospatial attribute-guided text enhancement (GATE) and hierarchical correlation recalibration (HCR), ultimately generating pixel-level segmentation predictions.

A. Image–Text Feature Extraction

Following the rotation-aware strategy introduced in OVRS [5], image features are extracted from multiple orientations. Consider an input image $I \in \mathbb{R}^{H\times W\times 3}$, where H and W denote the spatial dimensions and the last dimension corresponds to the spectral-band dimension. The image is rotated into multiple orientations, i.e., $0°$, $90°$, $180°$, and $270°$, to obtain rotated views $\{I^r\}_{r=0}^{R-1}$, where r indexes the rotated view and R denotes the number of orientations. These images are then fed into the CLIP image encoder $E_v(\cdot)$ to extract orientation-specific image features:

$$F_v^r = E_v(I^r), \tag{1}$$

where $F_v^r \in \mathbb{R}^{(H'W')\times D}$ denotes the flattened dense image features of the r-th orientation, D is the feature dimension, and $H' = H/s$ and $W' = W/s$ are the spatial dimensions of the feature map, with s denoting the patch size. These image features are concatenated in a single dimension to obtain the multi-orientation image features $F_v^{\mathrm{rot}} \in \mathbb{R}^{(H'W')\times R\times D}$:

$$F_v^{\mathrm{rot}} = \left[F_v^0, F_v^1, \ldots, F_v^{R-1}\right]. \tag{2}$$

The original-view CLIP image encoder jointly extracts the final-layer image feature F_v^0 and a set of intermediate image features $\mathcal{F}_{v,\mathrm{mid}}^0 = \{F_{v,u}^0\}_{u=1}^{N_u}$ from selected intermediate layers. These intermediate features are retained for progressive decoder fusion, where u indexes the selected intermediate layer and N_u denotes the total number of retained intermediate layers.

For the text branch, each candidate class is represented by a textual expression τ_c, e.g., *"an aerial view of [CLASS]"*. The CLIP text encoder $E_t(\cdot)$ takes the set of textual expressions $\mathcal{T} = \{\tau_c\}_{c=1}^{C}$ as input and extracts the text features F_t:

$$F_t = E_t(\mathcal{T}) \in \mathbb{R}^{C\times D}, \tag{3}$$

where C denotes the number of classes and D denotes the feature dimension.

B. Geospatial Attribute-Guided Text Enhancement

As illustrated in Fig. 3(b), GATE leverages a shared attribute prototype space and performs visual-context-guided attribute composition to generate adaptive and fine-grained textual representations, thereby improving semantic discrimination among visually similar land-cover categories.

1) Attribute Prototype Space Learning: Text representations derived from category names are coarse-grained and insufficient to capture the diverse attributes of remote sensing categories. Therefore, GATE organizes attribute descriptions into a shared attribute prototype space to reduce semantic redundancy and refines prototype representations to enhance semantic discrimination.

A large language model is employed to generate attribute-rich descriptions of each remote sensing class from an overhead-view perspective, including cues related to shape, texture, scale, surface appearance, and spatial context (the detailed LLM prompt is provided in Appendix A). These descriptions are encoded by the CLIP text encoder into attribute embeddings $\mathcal{Z}$. K-means clustering is applied to $\mathcal{Z}$ to construct the initial attribute prototype bank:

$$A^{(0)} = \mathrm{KMeans}(\mathcal{Z}, M) = \{a_m^{(0)}\}_{m=1}^{M}, \tag{4}$$

where M is set to the candidate value achieving the highest average silhouette width [35], and $a_m^{(0)} \in \mathbb{R}^D$ denotes the centroid of the m-th cluster.

The prototype bank $A = \{a_m\}_{m=1}^{M} \in \mathbb{R}^{D\times M}$ is initialized from $A^{(0)}$ and optimized using an InfoNCE-inspired [36] prototype separation loss:

$$\mathcal{L}_{\mathrm{attr}} = -\frac{1}{M}\sum_{i=1}^{M}\log\frac{\exp(a_i^\top a_i/\tau)}{\sum_{j=1}^{M}\exp(a_i^\top a_j/\tau)}, \tag{5}$$

where τ denotes the temperature parameter. After offline optimization, the refined prototype bank is frozen during segmentation training.

2) Visual-Context-Guided Attribute Composition: Attribute prototypes are dynamically selected and composed under the guidance of category-level semantic priors and image-specific visual contexts, generating image-adaptive attribute representations. Specifically, the relevance between category representations and attribute prototypes is computed as:

$$T = F_t A, \tag{6}$$

where $T \in \mathbb{R}^{C\times M}$ denotes the class–attribute compatibility matrix.

To incorporate image-specific visual contexts, the model computes attention weights between attribute prototypes and the input image, generating dynamic weights $w_a \in \mathbb{R}^{1\times M}$. The attribute composition weights P are then obtained as:

$$P = \mathrm{softmax}\left(\mathrm{Top}_k\left(T \odot w_a\right)\right), \tag{7}$$

where $P \in \mathbb{R}^{C\times M}$ represents the attribute composition distribution. The Top_k operation retains the most relevant attribute prototypes while suppressing redundant semantic responses.

The attribute prototypes are weighted by P and residually injected into the original text features F_t, yielding the enhanced text features F_t':

$$F_t' = PA^\top + F_t. \tag{8}$$

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C. Image–Text Correlation Construction

Following the rotation-aggregative similarity computation in OVRS [5], the initial image–text correlation maps S are computed using cosine similarity between the multi-orientation image features and text features:

$$S = \mathrm{CosSim}\left(F_v^{\mathrm{rot}}, F_t'\right), \quad S \in \mathbb{R}^{(H'W')\times R\times C}, \tag{9}$$

where $\mathrm{CosSim}(\cdot,\cdot)$ denotes cosine similarity. For each class $c \in \{1,\ldots,C\}$ and orientation $r \in \{0,\ldots,R-1\}$, the corresponding slice of S is reshaped into a spatial correlation map $S_c^r \in \mathbb{R}^{H'\times W'\times 1}$. In the following, the subscript c indicates that the corresponding quantity is associated with class c. The map S_c^r is then projected into a high-dimensional correlation embedding E_c^r for subsequent recalibration:

$$E_c^r = \mathrm{Conv}(S_c^r) \in \mathbb{R}^{H'\times W'\times d_F}, \tag{10}$$

where $Conv(\cdot)$ denotes a convolutional layer and d_F is the correlation embedding dimension. The embedding is then rotated back to the original orientation and denoted as $\bar{E}_c^r$. The embeddings $\{\bar{E}_c^r\}_{r=0}^{R-1}$ are concatenated and further fused by a convolution:

$$E_c = \mathrm{Conv}\left([\bar{E}_c^0, \bar{E}_c^1, \ldots, \bar{E}_c^{R-1}]\right) \in \mathbb{R}^{H'\times W'\times d_F}, \tag{11}$$

where $[\cdot]$ denotes concatenation along the orientation dimension and E_c denotes the correlation embedding fused across all orientations. The embeddings of all categories are collected as $E = \{E_c\}_{c=1}^{C} \in \mathbb{R}^{H'\times W'\times C\times d_F}$.

D. Hierarchical Correlation Recalibration

HCR refines image–text correlation embeddings through three complementary stages motivated by spatial inconsistency, feature redundancy, and category ambiguity. Spatial recalibration uses visual guidance to modulate correlation embeddings and model spatial dependencies. Feature recalibration derives channel weights from the intermediate correlation responses, while category recalibration incorporates GATE-enhanced textual guidance to model interactions among category-specific responses. These stages share the objective of improving correspondence discrimination through adaptive modulation, reweighting, and refinement of the correlation representation. The overall pipeline is illustrated in Fig. 3(c).

1) Spatial Reliability Recalibration: Due to visual similarities and spatial discontinuities, pixel-wise matching responses suffer from neighborhood interference, causing feature leakage and boundary blurring. To enforce spatial consistency, HCR introduces an image-conditioned spatial reliability recalibration mechanism. Specifically, it extends the FiLM [37] paradigm to the correlation space, leveraging visual features to adaptively recalibrate the correlation embedding E_c as:

$$\hat{E}_c = E_c \odot (1+\gamma_c) + \beta_c, \tag{12}$$

where $\gamma_c, \beta_c \in \mathbb{R}^{H'\times W'\times d_F}$ denote the spatially varying scale and shift maps derived from the visual feature F_v^0 via linear projection.

To complement this modulation with global context, a four-directional Mamba2 [38] scanning strategy is employed to establish long-range spatial dependencies within the modulated representation:

$$Y_c = \sum_{k=1}^{4} \Omega_k^{-1}(\mathrm{Mamba2}(\Omega_k(\hat{E}_c))), \tag{13}$$

where $\Omega_k(\cdot)$ denotes scanning along the k-th direction, and $\Omega_k^{-1}(\cdot)$ restores the scanned sequence to its original spatial layout. A lightweight gated MLP is further applied for local refinement to preserve spatial structures, producing the spatially recalibrated representation E_c^{spa}.

2) Feature Reliability Recalibration: Correlation embeddings may contain redundant or weakly informative feature channels. HCR therefore introduces feature-wise recalibration to adaptively reweight these channels using weights derived from efficient cross-channel interactions [39]:

$$w_c = \sigma(\mathrm{Conv}(\mathrm{GAP}(E_c^{\mathrm{spa}}))), \tag{14}$$

where $\mathrm{GAP}(\cdot)$ denotes global average pooling over spatial dimensions, $\mathrm{Conv}(\cdot)$ captures local feature interactions, and $\sigma(\cdot)$ represents the sigmoid function.

The feature reliability weights are applied to recalibrate the correlation embedding:

$$E_c^{\mathrm{feat}}(i,j,:) = E_c^{\mathrm{spa}}(i,j,:) \odot w_c, \tag{15}$$

where w_c denotes the feature-wise recalibration weights.

3) Category Reliability Recalibration: Visually similar categories frequently respond to the same visual evidence, leading to severe semantic competition. To resolve this inter-class ambiguity, HCR explicitly models category-level competition. By recalibrating responses across the category dimension, it models interactions among category-specific responses to improve category discrimination.

To model these interactions per pixel, the recalibrated representation at the p-th location is formulated as $E_p^{\mathrm{feat}} \in \mathbb{R}^{C\times d_F}$, where each row denotes a category-specific hypothesis. The competition among these hypotheses is then resolved by a cross-attention mechanism, leveraging the enhanced text representation F_t' as category-level semantic guidance:

$$\tilde{E}_p^{\mathrm{cat}} = E_p^{\mathrm{feat}} + \mathrm{Attn}(E_p^{\mathrm{feat}}, F_t'), \tag{16}$$

where $\mathrm{Attn}(\cdot,\cdot)$ denotes the attention operation between category-wise correlation embeddings and enhanced textual representations. The refined representation is further processed by an FFN and reshaped into $E^{\mathrm{cat}} \in \mathbb{R}^{H'\times W'\times C\times d_F}$, thereby suppressing spurious matches and yielding highly discriminative category-wise correlation representations.

E. Decoder

Since the correlation embeddings are generated at a low spatial resolution, they may lack fine-grained spatial details. To recover detailed spatial structures, the decoder progressively upsamples the category recalibrated correlation embeddings and integrates the retained intermediate image feature set

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$\mathcal{F}^0_{v,\mathrm{mid}}$. At the u-th decoding stage, the corresponding intermediate image feature $F^0_{v,u} \in \mathcal{F}^0_{v,\mathrm{mid}}$ is fused with the upsampled decoder feature:

$$E^{\mathrm{dec}} \leftarrow \mathrm{Proj}\left(\left[\mathrm{Up}\left(E^{\mathrm{dec}}\right); F^0_{v,u}\right]\right), \quad u = 1, \ldots, N_u, \quad (17)$$

where E^{dec} is initialized with E^{cat}, $\mathrm{Up}(\cdot)$ denotes upsampling, and $\mathrm{Proj}(\cdot)$ denotes feature projection and fusion. After N_u decoding stages, the decoder generates the final prediction $O \in \mathbb{R}^{H\times W\times C}$ at the original image resolution.

F. Training Objective

We use binary cross-entropy (BCE) loss to train our model. Given a one-hot segmentation mask $M \in \{0,1\}^{H\times W\times C}$ and the predicted logits $O \in \mathbb{R}^{H\times W\times C}$, the loss is defined as follows:

$$\mathcal{L}_{\mathrm{bce}} = \mathrm{BCE}\left(\sigma(O), M\right), \quad (18)$$

where $\sigma(\cdot)$ denotes the sigmoid function and $\mathrm{BCE}(\cdot,\cdot)$ denotes the binary cross-entropy loss. Minimizing this loss encourages accurate pixel-level class predictions.

IV. EXPERIMENTS

A. Datasets

Following the evaluation protocol of OVRSISBench [13], we train all methods on DLRSD [40] and evaluate them on seven reformulated remote sensing segmentation datasets, including iSAID [41], LoveDA [42], Potsdam, Vaihingen, UAVid [43], UDD5 [44], and VDD [45]. DLRSD is used as the training dataset, while all target datasets are excluded from training and used exclusively for open-vocabulary evaluation on both seen and unseen classes.

B. Evaluation Metrics

Four widely used evaluation metrics are adopted for a comprehensive assessment of OVRSIS performance. Specifically, mean Intersection over Union (mIoU), frequency-weighted Intersection over Union (fwIoU), mean accuracy (mACC), and pixel accuracy (pACC) are employed for evaluation [14], [46]. The definitions of these metrics are given as follows:

$$\mathrm{mIoU} = \frac{1}{C}\sum_{c=1}^{C}\frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FP}_c + \mathrm{FN}_c}, \quad (19)$$

$$\mathrm{fwIoU} = \frac{1}{\sum_{c=1}^{C} N_c}\sum_{c=1}^{C} N_c \frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FP}_c + \mathrm{FN}_c}, \quad (20)$$

$$\mathrm{mACC} = \frac{1}{C}\sum_{c=1}^{C}\frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FN}_c}, \quad (21)$$

$$\mathrm{pACC} = \frac{\sum_{c=1}^{C}\mathrm{TP}_c}{\sum_{c=1}^{C} N_c}, \quad (22)$$

where TP_c, FP_c, and FN_c denote the numbers of true-positive, false-positive, and false-negative pixels for class c, respectively. N_c denotes the number of ground-truth pixels belonging to class c, and C is the total number of evaluated classes.

C. Implementation Details

GeoFGA is implemented using PyTorch [48] and Detectron2 [49]. All experiments are conducted on a server equipped with two NVIDIA RTX 3090 GPUs, each with 24 GB of memory. We adopt CLIP with a ViT-B/16 image encoder and extract features from $R = 4$ image orientations. For decoder fusion, we retain $N_u = 2$ intermediate feature maps from the 3rd and 7th transformer blocks of the image encoder processing the original view. The correlation embedding dimension d_F is set to 128, and the number of HCR modules N_r is set to 1 by default. The AdamW [50] optimizer is used with a learning rate of 2×10^{-4} for the proposed model and 2×10^{-6} for fine-tuning the CLIP backbone, with the weight decay set to 10^{-4}. The batch size is set to 8. The model is trained for 30k iterations.

The GATE module generates $n = 10$ geospatial attribute descriptions for each class using GPT-4. The temperature parameter τ for prototype-level contrastive refinement is set to 0.07, while visual–attribute attention employs 8 heads with a Top_k retention ratio of 0.9. In HCR, the Mamba2-based spatial recalibration adopts a state dimension of 64, an expansion ratio of 2, and a head dimension of 32, whereas category recalibration employs 8 attention heads. For the 128-dimensional correlation features, Mamba2 uses 8 SSM heads. A single Mamba2 module processes the four sequences obtained from forward and reverse row-major and column-major scans with shared parameters. The directional outputs are restored to the original spatial layout and summed. Local refinement employs a gated MLP with an expansion ratio of 3, followed by a 7×7 depth-wise convolution.

D. Quantitative Results

Table I reports the quantitative comparisons on OVRSISBench [13], with DLRSD as the training dataset. We compare GeoFGA with representative general OVSS methods, including SCAN [28], SAN [47], SED [29], and CAT-Seg [7], as well as recent OVRSIS methods, including TPOV-Seg [8], OVRS [5], GSNet [9], and RSKT-Seg [13]. All compared methods adopt CLIP-based vision–language models for image–text alignment. For recent methods equipped with additional remote sensing priors, such as GSNet and RSKT-Seg, we report both their original results and CLIP-only results (denoted with * in Table I).

The results show that GeoFGA achieves the best overall performance, with an average mIoU of 40.12% and an average mACC of 56.10% across the seven evaluation datasets. Compared with the strongest competitor, RSKT-Seg, GeoFGA improves the average mIoU and average mACC by 2.90 and 1.65 percentage points, respectively. Compared with CAT-Seg, a general OVSS method, GeoFGA improves the average mIoU from 29.58% to 40.12%, demonstrating the effectiveness of remote-sensing-specific semantic enhancement.

GeoFGA also performs consistently across individual datasets, achieving the best mIoU on six datasets and the best mACC on six datasets. Notably, it surpasses RSKT-Seg by 7.08 and 4.74 percentage points in mIoU on LoveDA and Potsdam, respectively. These quantitative results demonstrate

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TABLE I
QUANTITATIVE COMPARISONS ON OVRSISBENCH WITH DLRSD AS THE TRAINING DATASET. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINED, RESPECTIVELY.

Method	Pretrained Models	Backbone	Type	iSAID		LoveDA		Potsdam		UAVid		UDD5		Vaihingen		VDD		Average	
				mIoU	mACC	mIoU	mACC	mIoU	mACC	mIoU	mACC	mIoU	mACC	mIoU	mACC	mIoU	mACC	mIoU	mACC
SCAN [28]	CLIP	ViT-B	OVSS	34.18	49.05	18.23	40.12	20.22	34.70	18.56	30.31	32.12	40.45	5.38	22.54	26.25	43.67	22.13	37.26
SAN [47]	CLIP	ViT-B	OVSS	30.63	44.03	23.15	48.26	30.30	44.98	22.34	37.56	36.87	47.21	31.92	45.36	34.76	52.42	30.00	45.69
SED [29]	CLIP	ConvNeXt-B	OVSS	21.54	36.28	21.32	45.17	19.47	33.40	20.12	33.34	34.65	44.10	29.40	49.38	31.43	50.25	25.42	41.70
CAT-Seg [7]	CLIP	ViT-B	OVSS	23.56	38.48	25.45	50.32	26.79	44.72	24.56	39.20	38.23	49.56	32.32	49.65	36.18	54.30	29.58	46.60
TPOV-Seg* [8]	CLIP	ViT-B	OVRSIS	38.27	57.34	17.70	41.44	25.32	48.42	23.53	36.75	39.47	55.76	23.20	43.08	34.91	54.79	28.91	48.23
OVRS [5]	CLIP	ViT-B	OVRSIS	39.09	54.43	28.67	52.10	27.47	42.07	25.23	40.18	39.10	50.65	33.71	49.01	37.34	55.15	32.94	49.08
GSNet* [9]	CLIP	ViT-B	OVRSIS	39.75	56.94	35.07	57.19	32.69	47.39	25.03	38.02	30.62	45.60	31.25	47.77	34.73	50.22	32.73	49.02
GSNet [9]	CLIP+DINO_{RS}	ViT-B	OVRSIS	42.00	59.19	29.32	53.02	26.46	43.20	25.42	40.70	40.05	51.72	35.15	52.62	38.10	56.01	33.79	50.92
RSKT-Seg* [13]	CLIP	ViT-B	OVRSIS	39.08	56.56	32.02	54.05	31.93	46.70	25.32	38.16	38.33	56.44	32.18	46.84	40.71	57.15	34.22	50.84
RSKT-Seg [13]	CLIP+DINO_{RS} +CLIP_{RS}	ViT-B	OVRSIS	**44.04**	61.23	32.49	55.67	34.53	50.71	28.14	42.86	42.99	57.81	37.16	**54.81**	41.22	58.09	37.22	54.45
GeoFGA	CLIP	ViT-B	OVRSIS	43.92	**62.27**	**39.57**	**60.32**	**39.27**	**53.17**	**31.12**	**43.89**	**43.51**	**58.33**	**37.98**	53.21	**45.46**	**61.53**	**40.12**	**56.10**

* denotes results reproduced by us using the official implementation; the subscript $_{RS}$ denotes remote sensing pre-training.

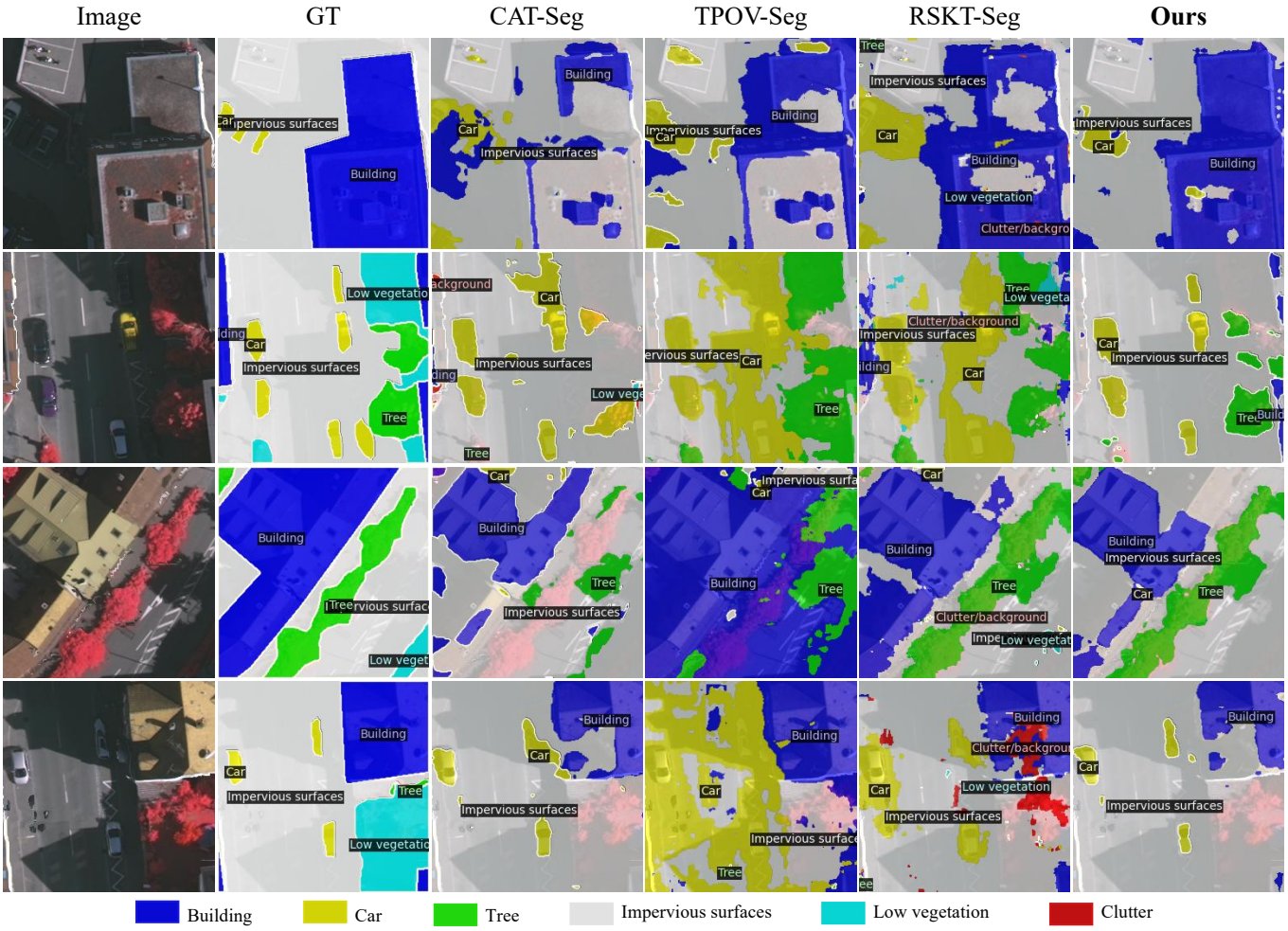


Fig. 4. Qualitative segmentation results on the Vaihingen dataset. From left to right, the columns show the input image, ground truth, CAT-Seg, TPOV-Seg, RSKT-Seg, and GeoFGA. Compared with existing methods, GeoFGA produces more complete building regions and reduces category confusion in complex adjacent land-cover areas.

the effectiveness and consistent performance of GeoFGA under open-vocabulary evaluation across diverse remote sensing datasets.

E. Visual Comparison

Fig. 4 presents qualitative comparisons on the Vaihingen dataset, an aerial urban dataset containing near-infrared, red, and green (IRRG) bands. Relative to the class vocabulary of DLRSD used for training, Vaihingen contains three seen and three unseen classes, providing a balanced setting for

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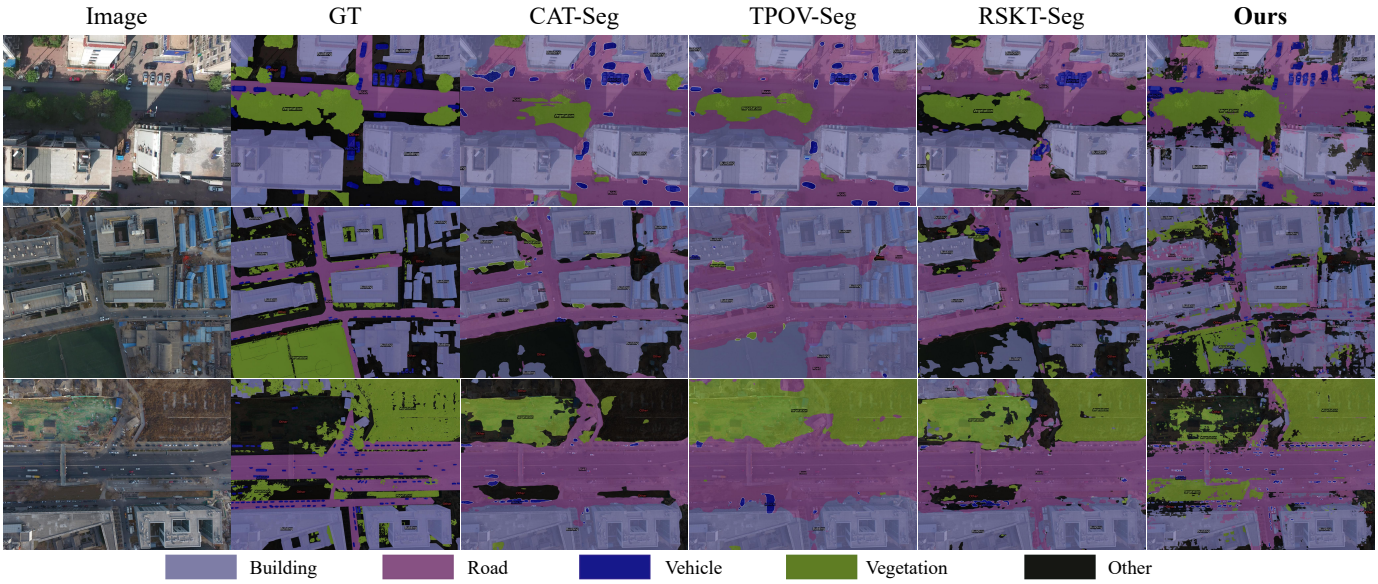


Fig. 5. Qualitative segmentation results on the UDD5 dataset. From left to right, the columns show the input image, ground truth, CAT-Seg, TPOV-Seg, RSKT-Seg, and GeoFGA. GeoFGA better preserves small objects and improves the discrimination between vegetation and surrounding background regions.

evaluating open-vocabulary segmentation. Its fine-grained urban classes and adjacent land-cover regions pose challenges for class discrimination and boundary delineation. CAT-Seg produces fragmented masks and inaccurate delineations for buildings and impervious surfaces, while TPOV-Seg tends to generate over-expanded masks, particularly for cars, buildings, and trees. RSKT-Seg further exhibits class confusion, producing incomplete building regions and spurious predictions for cars and clutter/background. In contrast, GeoFGA generates more coherent masks and better separates buildings, cars, trees, and impervious surfaces from neighboring regions. It also preserves compact car regions while suppressing nearby false activations, indicating improved fine-grained localization under the open-vocabulary setting. Nevertheless, some errors remain in vegetation regions, where several low-vegetation areas are misclassified as tree or other land-cover classes. These examples indicate that, although GeoFGA improves localization and reduces false activations, fine-grained discrimination among visually similar categories (e.g., low vegetation vs. tree in IRRG imagery) remains challenging.

Fig. 5 further presents qualitative results on UDD5, a UAV-based dataset characterized by small objects, complex backgrounds, and large variations in apparent object scale. The main challenge lies in distinguishing background regions from buildings, roads, and vegetation. CAT-Seg produces unstable predictions around vegetation and background areas, whereas TPOV-Seg and RSKT-Seg frequently confuse background with buildings or vegetation. Vehicles are also often missed or merged into connected regions rather than being identified as separate objects. By contrast, GeoFGA better preserves the semantic layout of roads, buildings, vegetation, and background regions, while producing more complete and compact predictions for vehicles and other foreground objects. These observations indicate improved small-object segmentation and reduced background interference in UAV imagery.

Overall, the qualitative comparisons show that GeoFGA produces more spatially coherent and class-consistent segmentation results, with more accurate target delineation and fewer erroneous predictions in surrounding regions.

F. Ablation Study

For ablation analysis, we adopt the default settings and report the average performance across the seven target datasets unless otherwise specified.

1) Effectiveness of Key Components: We investigate the contributions of the two key components in GeoFGA, i.e., GATE and HCR, with the results reported in Table II. The baseline model achieves an average mIoU of 32.00%. Introducing GATE improves the average mIoU to 34.79%, demonstrating that attribute-guided representations provide more discriminative textual guidance. However, since GATE primarily enhances textual semantics, the reliability of intrinsic image–text correlations requires further modeling. Applying HCR independently to the baseline improves the average mIoU to 36.96%. By directly operating on dense image–text correlations and improving matching reliability through recalibration, HCR yields a larger performance gain. Combining GATE and HCR yields an average mIoU of 40.12%, improving upon the baseline by 8.12 percentage points and exceeding the performance of GATE alone and HCR alone by 5.33 and 3.16 percentage points, respectively. These results confirm the synergy between the two modules. Specifically, GATE boosts category discrimination via image-adaptive textual representations, improving initial image–text correlations to guide HCR. HCR then refines these correlations by mitigating spatial inconsistency, feature redundancy, and inter-class ambiguity. Nevertheless, performance gains vary across datasets: on the oblique-view UAVid dataset, the full model achieves 31.12% mIoU, slightly below HCR alone (31.36%). This decrease

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TABLE II
ABLATION STUDY OF THE PROPOSED COMPONENTS USING MIOU (%).

GATE	HCR	iSAID	LoveDA	Potsdam	UAVid	UDD5	Vaihingen	VDD	Average
×	×	33.40	31.37	28.15	25.38	34.78	34.98	35.95	32.00
✓	×	32.98	32.37	33.25	29.82	40.28	38.36	36.49	34.79
×	✓	40.72	35.16	34.90	31.36	38.96	37.93	39.71	36.96
✓	✓	43.92	39.57	39.27	31.12	43.51	37.98	45.46	**40.12**

TABLE III
CATEGORY-LEVEL COMPARISON FROM DLRSD TO POTSDAM.

Methods	Impervious surfaces*		Building		Low vegetation*		Tree		Car		Clutter*		Total	
	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	mIoU	pACC
TPOV-Seg	22.23	31.65	53.47	63.58	37.15	**69.21**	**25.18**	**33.17**	13.90	**92.93**	0.00	0.00	25.32	48.12
RSKT-Seg	44.96	**90.98**	54.76	58.45	35.66	49.05	17.24	18.27	52.85	85.74	1.69	1.76	34.53	57.34
GeoFGA	**49.39**	85.47	**60.50**	**71.52**	**37.17**	50.62	23.34	28.50	**61.41**	78.58	**3.82**	**4.31**	**39.27**	**61.02**

* denotes classes unseen during training.

TABLE IV
CATEGORY-LEVEL COMPARISON FROM DLRSD TO VDD.

Methods	Other*		Wall*		Road*		Vegetation*		Vehicle*		Roof*		Water		Total	
	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	mIoU	pACC
TPOV-Seg	0.24	0.24	2.91	5.35	27.72	90.22	68.44	**87.21**	15.60	45.84	53.92	71.61	75.54	83.04	34.91	63.57
RSKT-Seg	**16.70**	**22.34**	0.70	0.88	30.34	86.26	71.58	81.06	30.00	49.93	**60.91**	**84.48**	78.31	81.71	41.22	**67.47**
GeoFGA	15.88	22.12	**3.88**	**13.06**	**39.05**	**90.71**	**75.23**	82.44	**47.55**	**67.81**	58.28	70.30	**78.34**	**84.24**	**45.46**	66.48

* denotes classes unseen during training.

TABLE V
ABLATION STUDY ON THE NUMBER OF STACKED HCR MODULES.

Number of HCR Modules (N_r)	mIoU	fwIoU	mACC
1	**40.12**	**48.41**	**56.10**
2	35.68	44.31	53.45
3	36.91	45.03	52.45
5	37.07	46.16	52.40

TABLE VI
COMPARISON WITH ALTERNATIVE TEXT ENHANCEMENT AND CORRELATION RECALIBRATION STRATEGIES.

Component	**Strategy**	**mIoU**
Text Enhancement	Prompt Tuning	32.71
	GATE	**34.79**
Correlation Recalibration	Aggregation (CAT-Seg)	34.98
	HCR	**36.96**

TABLE VII
ABLATION STUDY OF THE INTERNAL COMPONENTS OF GATE AND HCR.

Model Variants	mIoU	mACC
Component Analysis of GATE		
Full GATE	**34.79**	**53.60**
w/o Prototype Refinement	33.94	52.85
w/o Image-Attribute Relevance	33.76	52.85
w/o Top_k Filtering	33.53	52.57
Component Analysis of HCR		
Full HCR	**36.96**	**54.14**
w/o Spatial Reliability Recalibration	35.35	50.95
w/o Feature Reliability Recalibration	35.43	51.41
w/o Category Reliability Recalibration	32.92	51.33

TABLE VIII
COMPUTATIONAL EFFICIENCY COMPARISON.

Method	**Params (M)** ↓	**FLOPs (G)** ↓	**FPS** ↑	**Peak GPU Memory (GB)** ↓
CAT-Seg	**154.29**	<u>141.72</u>	13.963	<u>1.424</u>
TPOV-Seg	155.20	**123.00**	<u>18.575</u>	1.573
RSKT-Seg	400.56	605.10	8.409	2.261
GeoFGA	<u>154.64</u>	245.68	**21.796**	**0.963**

may reflect a mismatch between the overhead-view descriptions used to construct GATE's attribute prototypes (see Appendix A) and the oblique-view imagery, reducing the effectiveness of additional semantic guidance. Overall, stronger complementarity is expected when the attribute prototypes provide discriminative category cues and the initial pixel-to-category matching remains ambiguous.

2) Category-Level Generalization Analysis: We further analyze the category-level results on Potsdam and VDD to examine the generalization ability of GeoFGA across seen and unseen classes. Unseen classes refer to classes that are absent from the DLRSD training set and are marked with an asterisk in Tables III and IV.

As shown in Table III, GeoFGA achieves the best overall performance on Potsdam, with 39.27% mIoU and 61.02% pACC. It obtains competitive results on both seen categories, such as building and car, and unseen classes, such as impervious surfaces and low vegetation, indicating its ability

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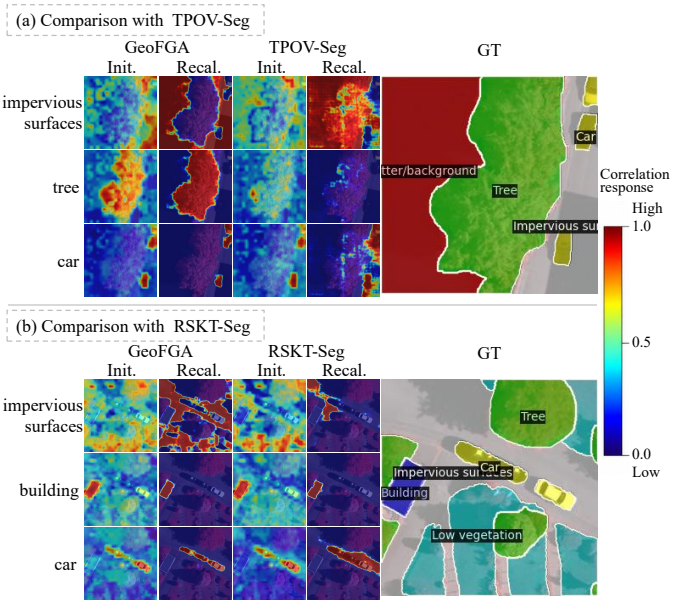


Fig. 6. Qualitative comparison of image–text correlation maps. (a) GeoFGA vs. TPOV-Seg. (b) GeoFGA vs. RSKT-Seg. Init. and Recal. denote the initial and recalibrated image–text correlation maps, respectively. Compared with existing methods, GeoFGA produces clearer and more spatially compact class responses, with fewer spurious activations in adjacent non-target regions. GT denotes the ground truth.

to adapt to category discrepancies between the source and target datasets. Notably, GeoFGA achieves 61.41% IoU on car, showing its advantage in localizing small-scale objects. Table IV reports the category-level results on VDD. GeoFGA achieves the best mIoU of 45.46%, outperforming TPOV-Seg and RSKT-Seg by 10.55 and 4.24 percentage points, respectively. It shows clear improvements on several unseen classes, including road, vegetation, and vehicle, suggesting that fine-grained image–text alignment facilitates transfer to novel category spaces. Although RSKT-Seg obtains a slightly higher pACC, GeoFGA achieves better class-balanced performance, as reflected by its higher mIoU. However, GeoFGA remains limited on certain ambiguous or structurally indistinct unseen categories, achieving only 3.82% IoU on clutter in Potsdam and 3.88% IoU on wall in VDD. The heterogeneous appearances of clutter may make it difficult for GATE to construct consistently discriminative attribute representations. HCR may also be limited when the initial correlations provide weak category-specific evidence.

3) Effect of the Number of HCR Modules: Table V analyzes the impact of stacking different numbers of HCR modules. The results show that the best overall performance is obtained when $N_r = 1$, achieving 40.12% mIoU, 48.41% fwIoU, and 56.10% mACC. Increasing N_r does not bring further gains and may introduce unnecessary refinement of the correlation embeddings. This indicates that a single HCR module can already provide effective spatial, feature, and category recalibration.

4) Comparison with Alternative Strategies: To fairly assess the effectiveness of the proposed designs, we evaluate each alternative strategy independently by incorporating it into the baseline model. For text enhancement, GATE is compared against CoOp-style Prompt Tuning [51], which optimizes learnable context tokens but lacks explicit geospatial attribute modeling and image-conditioned semantic composition. As shown in Table VI, GATE achieves an average mIoU of 34.79%, outperforming Prompt Tuning (32.71%). This result suggests that attribute-based semantic composition better captures fine-grained semantic information than learnable context embeddings alone. For correlation recalibration, HCR is compared with CAT-Seg's cost aggregation strategy. As shown in Table VI, HCR achieves a higher average mIoU than the aggregation strategy (36.96% vs. 34.98%). This performance gap suggests that contextual aggregation alone may be insufficient to fully resolve unreliable image–text correspondences in complex remote sensing scenes. By explicitly modeling correspondence reliability across the spatial, feature, and category dimensions, HCR more effectively suppresses ambiguous responses and improves dense image–text alignment. These results demonstrate the effectiveness of the proposed designs in addressing semantic alignment challenges in open-vocabulary remote sensing segmentation.

5) Ablation Study of Internal Components: Table VII reports the component-wise ablation results of GATE and HCR. We independently ablate the internal components of GATE and HCR on the baseline model, defining "Full GATE" and "Full HCR" as the baseline equipped with only the respective complete module. For GATE, removing prototype refinement decreases mIoU from 34.79% to 33.94%, indicating that refined prototypes provide more discriminative semantic representations. Removing image-attribute relevance reduces mIoU to 33.76%, highlighting the importance of image-conditioned attribute selection. Without Top_k filtering, the largest performance degradation is observed (33.53%), indicating that filtering irrelevant prototypes helps reduce semantic noise in complex scenes. For HCR, removing spatial, feature, and category reliability recalibration decreases mIoU by 1.61, 1.53, and 4.04 points compared with Full HCR (36.96%), respectively. The substantial degradation after removing category recalibration highlights the importance of resolving semantic competition among visually similar categories. Together, these complementary recalibration dimensions reduce correspondence ambiguity and improve dense image-text alignment.

6) Analysis of Model Efficiency: Table VIII compares the computational efficiency of the evaluated methods on a single NVIDIA GeForce RTX 3090 GPU using 384×384 inputs with a batch size of 1. Inference FPS is measured over 500 CUDA-synchronized inference iterations following 50 warm-up iterations, with host-to-device transfer included in the timed region. GPU memory denotes the peak allocated memory during inference. FLOPs are estimated using `fvcore`, supplemented by an analytical State Space Duality estimate for Mamba2 operators.

GeoFGA maintains a comparable parameter count of 154.64 M while achieving the highest inference throughput of 21.796 FPS and the lowest peak allocated GPU memory of 0.963 GB among the evaluated methods. Its peak allocated memory is 32.4% and 57.4% lower than that of CAT-Seg and RSKT-Seg, respectively. Although its estimated FLOP count of 245.68 G exceeds those of CAT-Seg (141.72 G) and

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TPOV-Seg (123.00 G), GeoFGA achieves higher measured inference throughput. This higher FLOP count partly results from encoding four orientation views. However, practical GPU throughput depends not only on arithmetic workload, but also on parallelizability, memory access cost, and kernel-launch overhead. Specifically, GeoFGA employs a single HCR module with batched four-directional scans via a fused Mamba2 execution path, whereas the evaluated CAT-Seg and TPOV-Seg configurations use two attention-based aggregation layers involving tensor rearrangements. These implementation differences can affect execution efficiency, so FLOP counts alone do not determine the measured FPS.

7) Visualization of Correlation Maps: Fig. 6 visualizes the effect of HCR on image–text correlation refinement. Benefiting from the semantic enhancement of GATE, the initial correlation maps of GeoFGA already provide relatively clear category localization. However, ambiguous responses and scattered activations still exist in semantically similar or spatially adjacent regions. After HCR, GeoFGA further produces more compact and reliable correlation responses by improving spatial consistency and feature discriminability while reducing inter-class ambiguity. Compared with TPOV-Seg and RSKT-Seg, GeoFGA more accurately highlights trees, cars, buildings, and impervious surfaces, while suppressing irrelevant responses in nearby regions. These results demonstrate that HCR effectively improves the reliability of dense image–text correspondence for segmentation prediction.

V. Conclusion

In this article, we address the challenge of unreliable image–text correspondence in open-vocabulary remote sensing semantic segmentation. To overcome the limitations of coarse textual semantics and ambiguous dense matching, we propose GeoFGA, a geospatial fine-grained alignment framework that enhances semantic representations and improves correspondence reliability. Specifically, GATE enriches class representations with fine-grained geospatial attributes by constructing a shared attribute prototype space and adaptively composing the prototypes within it, while HCR recalibrates ambiguous correlations through hierarchical reliability modeling across the spatial, feature, and category dimensions.

Extensive experiments across seven remote sensing benchmarks demonstrate the effectiveness of GeoFGA, which consistently outperforms existing methods in average performance and exhibits strong generalization across diverse category spaces. Comprehensive ablation studies and qualitative analyses further validate the overall contributions of GATE and HCR and the efficacy of reliability-oriented correlation refinement. Nevertheless, GeoFGA still faces challenges in distinguishing visually similar classes and recognizing ambiguous unseen categories with heterogeneous appearances. Its effectiveness also depends on the quality and coverage of the generated attribute knowledge. Future work will investigate more discriminative attribute generation and adaptive semantic composition to improve category discrimination and generalization to unseen classes.

Appendix A
LLM Prompt for Geospatial Attribute Generation

To ensure reproducibility, we provide the complete prompt used to generate fine-grained geospatial attribute descriptions for GATE. The prompt is designed to constrain the LLM to visually observable characteristics in overhead remote sensing imagery while excluding functional or non-visual semantic information.

Prompt:

Role: You are an expert in remote sensing image interpretation and computer vision.
Task: Generate a valid JSON object containing 10 fine-grained visual descriptions for each of the following remote sensing categories from a strict top-down aerial or satellite perspective.
Categories: [airplane, bare soil, buildings, cars, chaparral, court, dock, field, grass, mobile home, pavement, sand, sea, ship, tanks, trees, water]
Constraints:
1) **Observable visual attributes only.** Describe only characteristics that can be directly inferred from overhead imagery. Focus on discriminative cues such as geometric shape, scale, surface appearance, texture, reflectance or intensity patterns, shadow characteristics, spatial arrangement, and contrast with surrounding regions.
2) **No functional or encyclopedic semantics.** Do not describe object functions, purposes, historical information, human activities, or other knowledge that is not visually observable in overhead imagery.
3) **Remote sensing perspective.** All descriptions must reflect how the category typically appears under a top-down aerial or satellite view. Avoid ground-level or perspective-view characteristics that are not observable in overhead imagery.
4) **Fine-grained and discriminative descriptions.** Each description should emphasize visually informative attributes that can help distinguish the target category from other visually similar remote sensing categories. The 10 descriptions for each category should capture diverse visual characteristics and avoid unnecessary repetition.
5) **Description format.** Each description must follow a two-part format. The first part uses the template "An aerial image of [CLASS]." Here, [CLASS] denotes a grammatically appropriate noun phrase for the category, including an article where necessary. The second part must be a single sentence describing a specific visible attribute and beginning with "It" or "They", as grammatically appropriate. Keep the original category names unchanged as JSON keys.
6) **Terminology.** Use concise and professional remote sensing terminology where appropriate, such as spectral or intensity contrast, surface texture, orthogonal edges, specular reflection, shadow profile, spatial regularity, and apparent vertical relief. Do not introduce technical terms unless they describe

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visually observable characteristics.

Output format: Return only a valid JSON object. Each key must be the category name, and its value must be a list containing exactly 10 string descriptions. Do not include explanations, Markdown formatting, or any additional text outside the JSON object.

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